

REGAL v6

Complete Analysis Reference — 3-Tier Framework

SELLAS Galinpepimut-S (GPS) Phase 3 in AML CR2

Generated May 13, 2026 · Post-78-event anchor · Leaky-cure family · Shape ≤ 1.0 throughout

v6 framework changes

Collapsed BEDROCK and BEDROCK+ into a single BEDROCK tier reflecting all priors that are well-supported biologically or by the literature: cure ≥ 0.40 , leak ≤ 0.10 , shape ≤ 1.0 , ratio ≥ 0.85 . The old framework treated unc_shape and ratio as 'soft' priors layered on top of a no-biology floor; v6 recognizes that both are sufficiently well-supported to belong at the floor.

Tightened shape cap from ≤ 1.2 to ≤ 1.0 across all tiers. The shape cap at 1.2 was theoretically permissive but practically identical to ≤ 1.0 in this dataset (the simulation's shape grid is coarse: only 0.5, 1.0, 1.5, 2.0 are populated). Using ≤ 1.0 honestly reflects what the cap actually does: excludes accelerating-hazard scenarios (1.5, 2.0) which have no biological mechanism for AML CR2.

Ratio progression across tiers: ≥ 0.85 (BEDROCK) $\rightarrow \geq 0.90$ (STRONG) $\rightarrow \geq 0.95$ (MODERATE). Captures the natural prior probability decomposition: ratio < 0.85 has $< 5\%$ prior mass (no plausible mechanism), ratio 0.85–1.0 has $\sim 20\%$ mass (allocation-asymmetry / salvage-delay scenarios), ratio ≥ 1.0 has $\sim 75\%$ mass (no-harm / matches vs-placebo vaccine pattern).

Quick reference

Metric	Value
Defensible range	84% (BEDROCK) – 96% (MODERATE)
Floor (BEDROCK)	84.3%
Literature-anchored (STRONG)	94.6%
Published commit (MODERATE)	96.0%
Single-number anchor (STRONG–MODERATE midpoint)	$\sim 95\%$
Modal observed HR at readout	0.37 – 0.44
Modal readout date	mid-July 2026
Readout window (P05–P95)	late June – mid August 2026
Events remaining	2 (78 confirmed May 11, 2026)
Implied GPS survival benefit	~ 15 – 21 months over BAT

Trial context

REGAL (NCT04229979) is a Phase 3 randomized open-label trial of galinpepimut-S (GPS) maintenance vs Best Available Therapy (BAT) in adult patients with AML in second complete remission (CR2) who are ineligible for or unable to undergo allogeneic stem cell transplant. n=126 randomized 1:1, enrolled 2020–April 2024. Primary endpoint is overall survival. Success threshold: observed HR < 0.636 at 80 events.

Hard public anchors

Anchor	Value	Source
Trial start	Feb 8, 2021 (t=0)	SAP / public timeline
60 events (IA)	Dec 2024, month 46	SLS PR Dec 2024
IA verdict	IDMC: continue (passed futility)	SLS PR Jan 2025
Median follow-up at IA	13.5 ± 2 months	SLS PR Jan 2025
Pool blinded mOS at IA	> 12 months	SLS PR Jan 2025
72 events	Dec 26, 2025, month 58	SLS PR Dec 2025
Second IDMC review	Aug 2025, continue	SLS PR Aug 2025
78 events	May 11, 2026, month 62.97	SLS PR May 12, 2026
T-cell IR rate disclosed	80% (P3)	SLS PR Jan 2025
Final readout trigger	80 events	SAP
Success HR threshold	observed HR < 0.636	SAP

Methodology

Framework: ABC (Approximate Bayesian Computation) fitting the leaky-cure family to public anchors. Leaky-cure model: cure_frac of GPS patients reach a plateau with small late-relapse hazard (leak_yr); (1 – cure_frac) follow a Weibull survival distribution with median unc_med and shape unc_shape.

Cox PH sampling noise: SE on log(HR) at 80 events $\approx 2/\sqrt{80} = 0.224$. Applied as multiplicative log-normal noise to each combo's parametric HR to compute realized observed HR distribution. All tables use Cox SE = 0.22.

Parameter definitions

Parameter	Meaning	Typical range
cure_frac	Fraction of GPS patients reaching long-term plateau	0.0 – 0.7
leak_yr	Annual hazard of late relapse for cured patients (/yr)	0.0 – 0.10
ratio	Uncured GPS median / BAT median (no-harm metric)	0.5 – 2.0
unc_shape	Weibull shape for uncured GPS survival	0.5 – 2.0
unc_med	Median OS of uncured GPS patients (months)	8 – 30
bat_med	Median OS of BAT arm (months)	8 – 25
median_hr_final	Central parametric HR at 80 events	0.30 – 0.70
median_t80_months	Calendar month when 80th event occurs	64 – 67

The three-tier framework

Parameter	BEDROCK	STRONG	MODERATE
cure_frac	≥ 0.40	≥ 0.45	≥ 0.50
leak_yr	≤ 0.10	≤ 0.07	≤ 0.05
unc_shape	≤ 1.0	≤ 1.0	≤ 1.0
ratio (unc/BAT)	≥ 0.85	≥ 0.90	≥ 0.95
n combos	1,100	243	98
P(success)	84.3%	94.6%	96.0%

Tier interpretations

BEDROCK. Firm conservative floor: everything I'm willing to defend without invoking optimistic biology. Includes the four constraints with strong literature/biology support at their permissive ends. This is the defensible 'lower bound' under reasonable assumptions, not the 'no biology at all' lower bound.

STRONG. Literature-anchored center. Each prior tightened to where the published evidence actually supports it. The single cleanest 'defensible center' tier.

MODERATE. Originally published mainstream commit. Tightest priors representing the optimistic-end of what the biological evidence supports. Sits at the upper bound of the defensible range.

Note on tier compression

STRONG (94.6%) and MODERATE (96.0%) are only 1.5 pp apart. This is the structurally correct answer, not a bug. The leak step from 0.10 to 0.07 (between BEDROCK and STRONG) captures most of the lift; further tightening to 0.05 (between STRONG and MODERATE) is incremental because leak's marginal effect diminishes as the cap tightens. The BEDROCK→STRONG step is the substantive prior-strengthening; STRONG→MODERATE refines rather than fundamentally re-anchors.

Evidence for each prior

cure_frac (firm anchor; very strong support for ≥ 0.40)

- 80% T-cell IR disclosed in Phase 3 (SLS PR Jan 2025), surpassing P2 rates
- GPS Phase 1/2 in AML CR2: median OS 21.0 months at FU 30.8 months vs 5.4 months historical ($p < 0.02$)
- P2 IR-to-plateau conversion ratio ≈ 0.73 ; $80\% \times 0.73 = \sim 58\%$ implied cure
- The 80% IR datapoint is a hard public anchor observable from outside the trial

unc_shape ≤ 1.0 (very strong support)

- AML CR2 mortality is heavily front-loaded (Schlenk, Döhner, Estey reviews)
- Among 942 AML relapses, only 11 (1.16%) relapsed after CR > 5 years (Verma 2010)
- 68% of post-allo-HSCT relapses occur in year 1
- Biologically: shape < 1 (decreasing hazard) is the expected pattern; shape = 1 (constant hazard) is permissive upper bound. Shape > 1 (accelerating hazard) has no known biological mechanism in this setting.
- Note: shape grid in simulation is coarse (values 0.5, 1.0, 1.5, 2.0); shape $\leq 1.0 \equiv$ shape ≤ 1.2 in practice

ratio progression (strong support across the range)

Prior probability decomposition over the true ratio (uncured GPS median / BAT median):

Range	Prior mass	Mechanism
ratio < 0.85	$\sim 5\%$	Requires GPS to materially worsen outcomes beyond delay — thin tail
ratio 0.85 – 1.00	$\sim 20\%$	Plausible allocation asymmetry: delayed salvage on GPS arm
ratio ≥ 1.00	$\sim 75\%$	No allocation asymmetry; matches vs-placebo vaccine pattern

Tier mapping: BEDROCK ratio ≥ 0.85 captures $\sim 95\%$ of prior mass; STRONG ratio ≥ 0.90 captures $\sim 85\%$; MODERATE ratio ≥ 0.95 captures $\sim 80\%$. Each step admits a tighter portion of the salvage-delay tail.

Direct evidence supporting ratio at or near 1.0:

- Vx-001 (TERT vaccine NSCLC): non-responders 13.4mo OS vs placebo 11.3mo — never worse than placebo
- OPT-822/821 (breast), pancreatic neoantigen vaccine (AACR 2026): same pattern; non-responders match placebo
- Mechanistically: vaccine non-response is immunological failure, not disease acceleration
- REGAL is vs-BAT (active treatment), not vs-placebo — introduces a salvage-delay mechanism not present in the placebo trials; this is why we permit ratio < 1.0 in BEDROCK and STRONG

leak_yr (MODERATE support; weakest of the four priors)

Pro: GPS Phase 2 plateau at 30 months follow-up. WT1 vaccine review across 9 trials shows long-term survivors > 8 years.

Con: P2 plateau is partially circular (it IS the evidence we cite, and also what we're trying to validate). Allo-HSCT comparator implies 2–4%/year late hazard for 'true cure' — vaccine should be weaker. OCV-501 (different WT1 vaccine in CR1) showed plateau in BOTH arms — plateau may be CR1 biology, not vaccine-caused.

Honest read: leak ≤ 0.05 is plausible but optimistic; leak ≤ 0.07 (allo-HSCT comparable) is the strong-evidence ceiling. Confidence: $\sim 95\%$ for ≤ 0.10 , $\sim 80\%$ for ≤ 0.07 , $\sim 55\%$ for ≤ 0.05 . This is the prior doing the most differentiation work across tiers.

Metrics tables

Table 1 — Headline metrics

Tier	n	P(success)	P(fail)	Mode HR	HR P50	HR P05	HR P95
BEDROCK	1,100	84.3%	15.7%	0.44	0.48	0.30	0.76
STRONG	243	94.6%	5.4%	0.40	0.42	0.27	0.64
MODERATE	98	96.0%	4.0%	0.37	0.41	0.27	0.62

Table 2 — HR outcome distribution (per-bin %, Cox SE=0.22)

HR range	BEDROCK	STRONG	MODERATE
< 0.30 spectacular	4.6%	9.9%	12.1%
0.30 – 0.35 very strong	8.4%	14.5%	16.4%
0.35 – 0.40 modal zone	12.9%	18.4%	19.5%
0.40 – 0.45 strong	15.2%	17.8%	17.7%
0.45 – 0.50 strong	14.8%	14.3%	13.5%
0.50 – 0.55 solid	12.8%	10.0%	9.0%
0.55 – 0.636 squeak	15.5%	9.6%	7.9%
≥ 0.636 FAIL	15.7%	5.4%	4.0%

Table 3 — Cumulative thresholds P(HR < x)

Threshold	BEDROCK	STRONG	MODERATE
P(HR < 0.30) spectacular blowout	4.6%	9.9%	12.1%
P(HR < 0.35) blockbuster	13.0%	24.5%	28.5%
P(HR < 0.40) P2-magnitude	26.0%	42.9%	48.0%
P(HR < 0.45) strong win	41.2%	60.7%	65.7%
P(HR < 0.50) strong win	56.0%	75.0%	79.2%
P(HR < 0.55) comfortable win	68.9%	85.0%	88.2%
P(HR < 0.636) SUCCESS	84.3%	94.6%	96.0%

Table 4 — Readout timing & alive counts

Metric	BEDROCK	STRONG	MODERATE
t80 P05 (months)	64.4	64.2	64.1
t80 P50 (calendar)	Jul 16	Jul 17	Jul 17
t80 P95 (months)	65.9	66.0	66.1
BAT alive @ m58 (P50)	18.0	17.0	17.0
GPS alive @ m58 (P50)	35.0	37.0	37.0
BAT:GPS alive ratio	1 : 1.94	1 : 2.18	1 : 2.18
Total alive today (m63)	48	48	48
Total alive at readout (m65)	46	46	46

Timing is essentially identical across tiers — the 78-event anchor dominates.

Table 5 — Implied biology (acceptance-weighted P50)

Parameter	BEDROCK	STRONG	MODERATE
BAT mOS P50 (months)	15.0	14.5	14.5
BAT mOS P05–P95	11.5 – 18.0	11.5 – 16.0	12.0 – 16.0
Cure fraction P50	0.40	0.50	0.50
Leak rate P50 (/yr)	0.080	0.060	0.040
Uncured shape P50	1.00	1.00	1.00
Uncured GPS mOS P50 (months)	20.0	18.0	18.0

Table 6 — Parametric (true) HR distribution

Metric	BEDROCK	STRONG	MODERATE
Parametric HR P05	0.364	0.339	0.329
Parametric HR P50	0.479	0.417	0.400
Parametric HR P95	0.639	0.523	0.508

The parametric distribution is what the model believes about the underlying TRUE HR, stripped of Cox sampling noise. Note that even BEDROCK's parametric P95 (0.64) is only modestly above the success threshold — most of the observed-HR failure tail is sampling noise, not 'the drug doesn't work.'

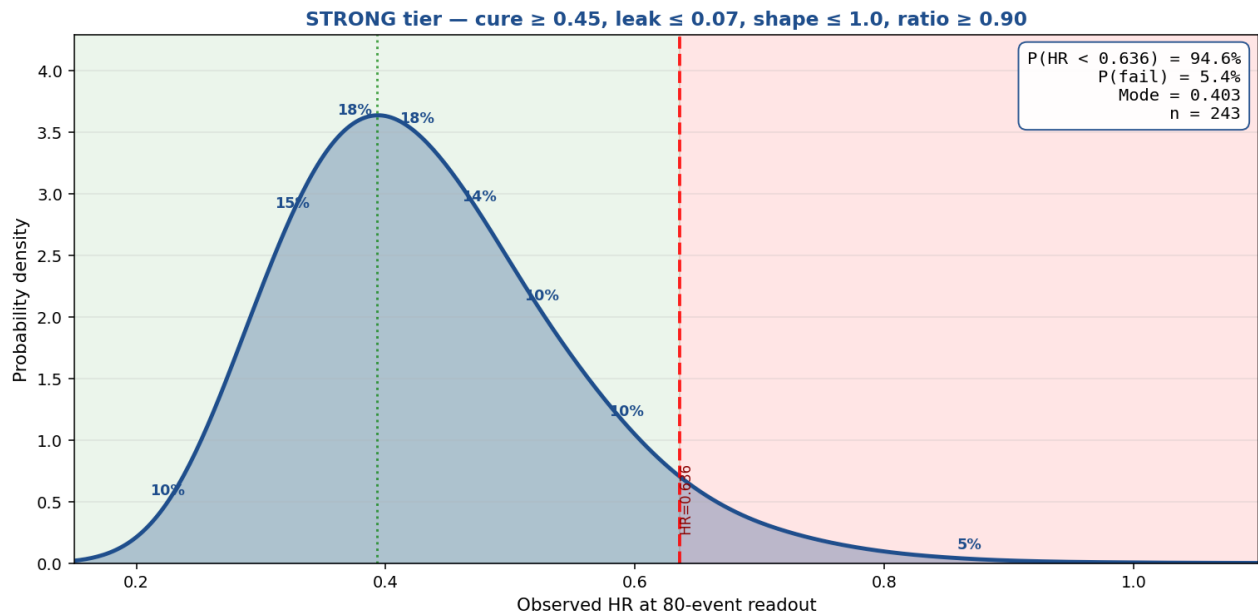
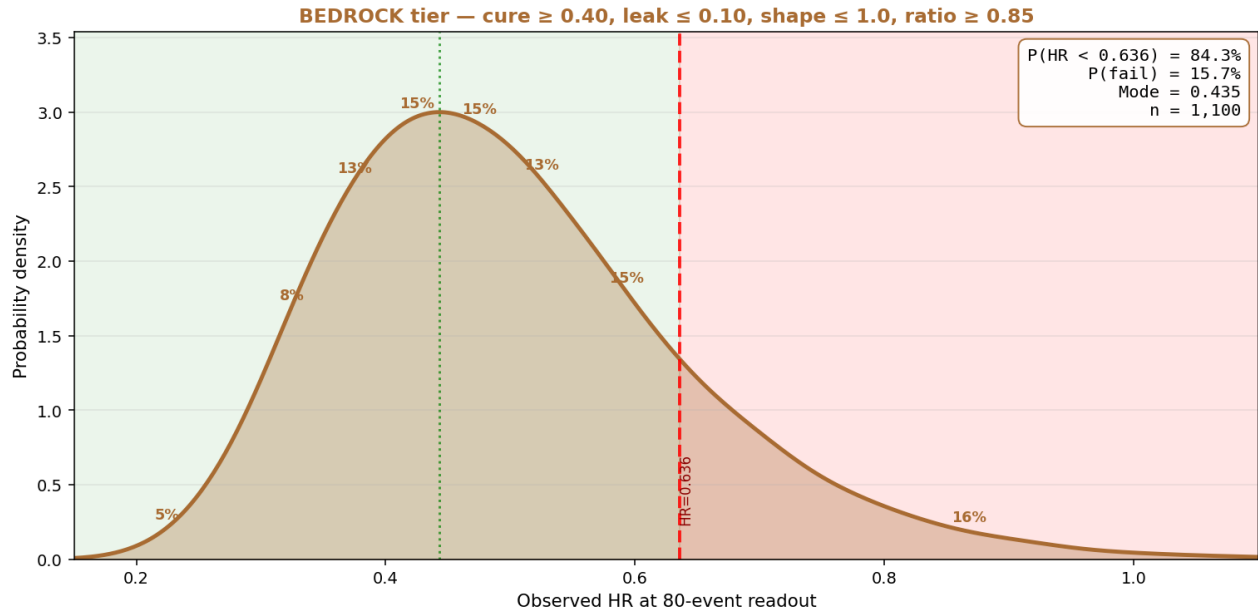
Table 7 — Failure tail decomposition

Source	BEDROCK	STRONG	MODERATE
P(fail) total observed (SE=0.22)	15.7%	5.4%	4.0%
P(fail) at SE=0.10 (bigger trial)	8.4%	0.5%	0.1%
P(fail) parametric only (no noise)	5.40%	0.00%	0.00%
P(fail) attributable to Cox noise	+10.3pp	+5.4pp	+4.0pp

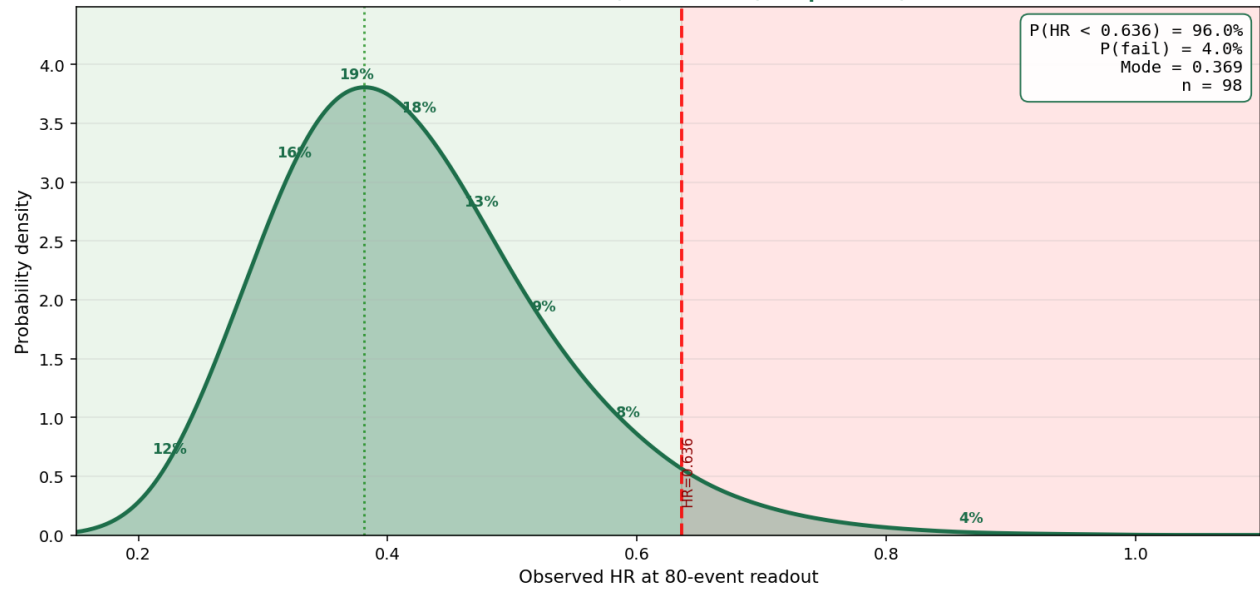
Cox sampling noise inflates BOTH tails of the observed HR distribution. Most of the failure mass is sampling-noise on near-threshold parametric scenarios, not 'the drug doesn't work.'

HR distribution charts

Posterior density of observed HR at 80-event readout, with Cox PH sampling noise (SE=0.22) applied. Green-tinted region = SUCCESS; red-tinted region = FAIL. Bin percentages along each curve sum to 100%. Mode (dotted green) is the peak of density.

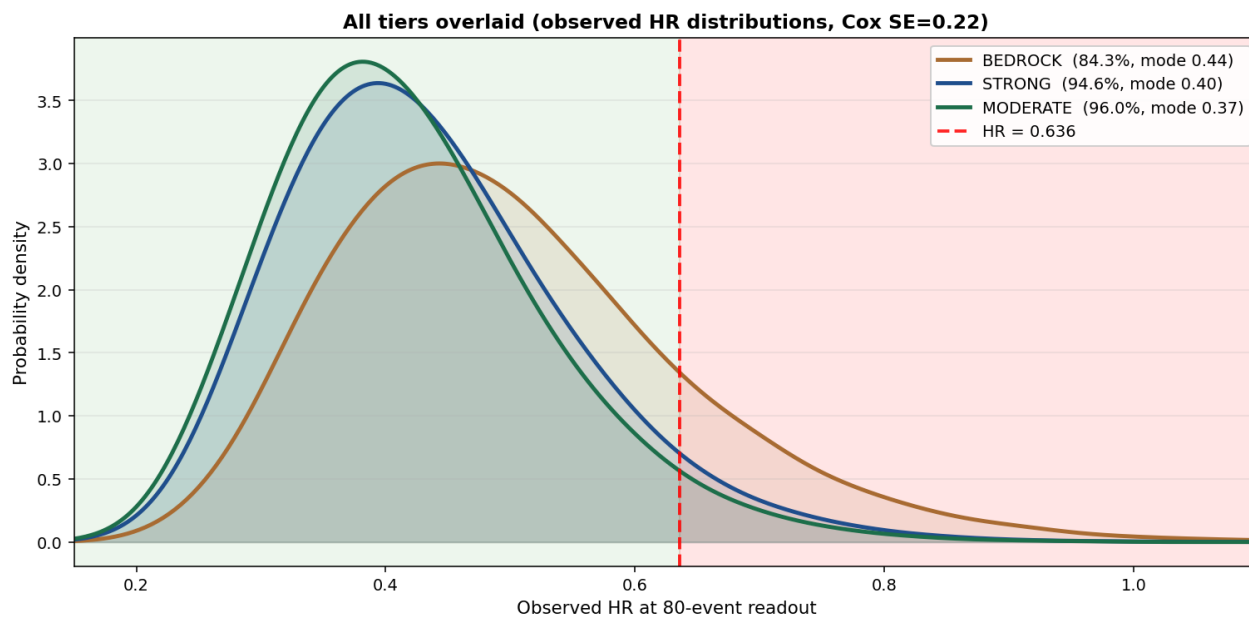


MODERATE tier — cure ≥ 0.50 , leak ≤ 0.05 , shape ≤ 1.0 , ratio ≥ 0.95



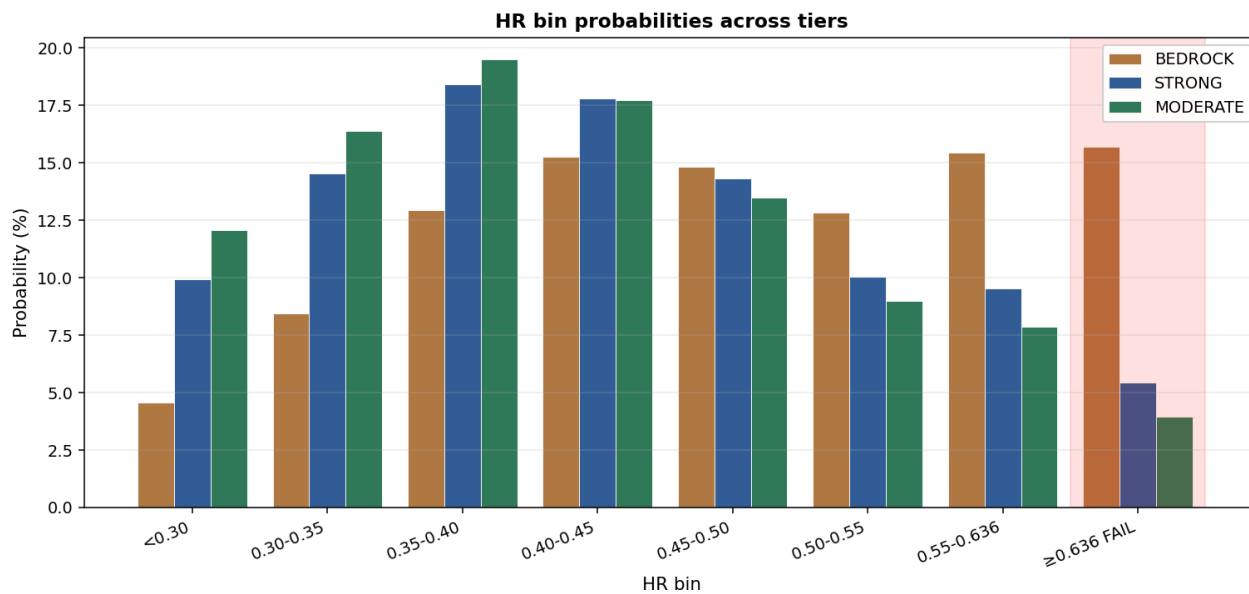
Tier comparison views

All tiers overlaid (observed HR)



The big density shift is between *BEDROCK* and *STRONG* (mode 0.44 → 0.40); *STRONG* and *MODERATE* are close.

HR bin probabilities side-by-side



Tail decomposition: parametric vs Cox sampling noise

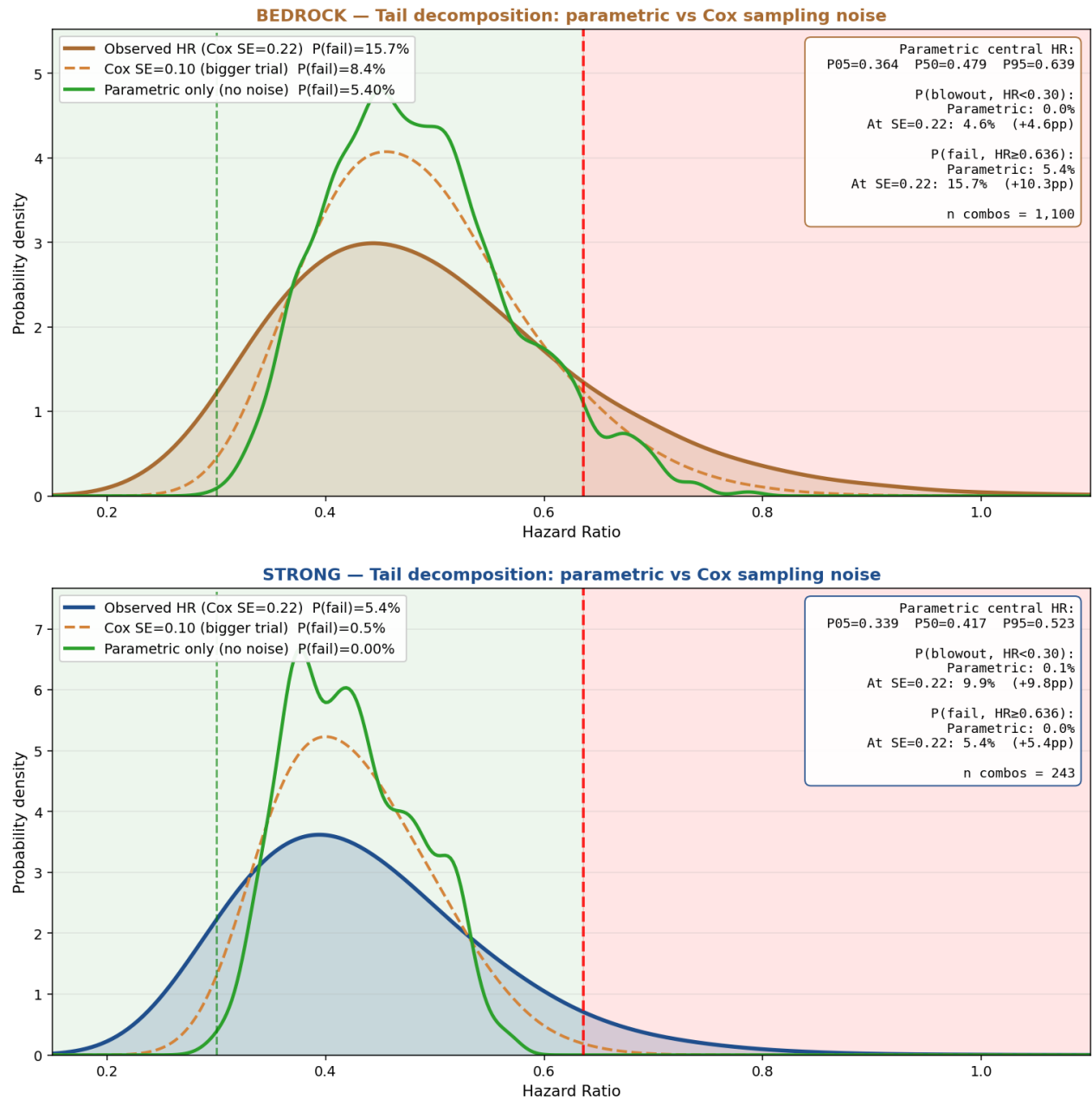
Key insight: Cox noise is bidirectional. At 80 events, Cox PH regression has SE on $\log(\text{HR}) \approx 0.22$. This sampling noise spreads the OBSERVED HR distribution in BOTH directions around the parametric (true) HR.

- **RIGHT tail (failure risk):** true HR = 0.50 can read as 0.65+ from bad luck
- **LEFT tail (blowout chance):** true HR = 0.50 can read as 0.35 from good luck

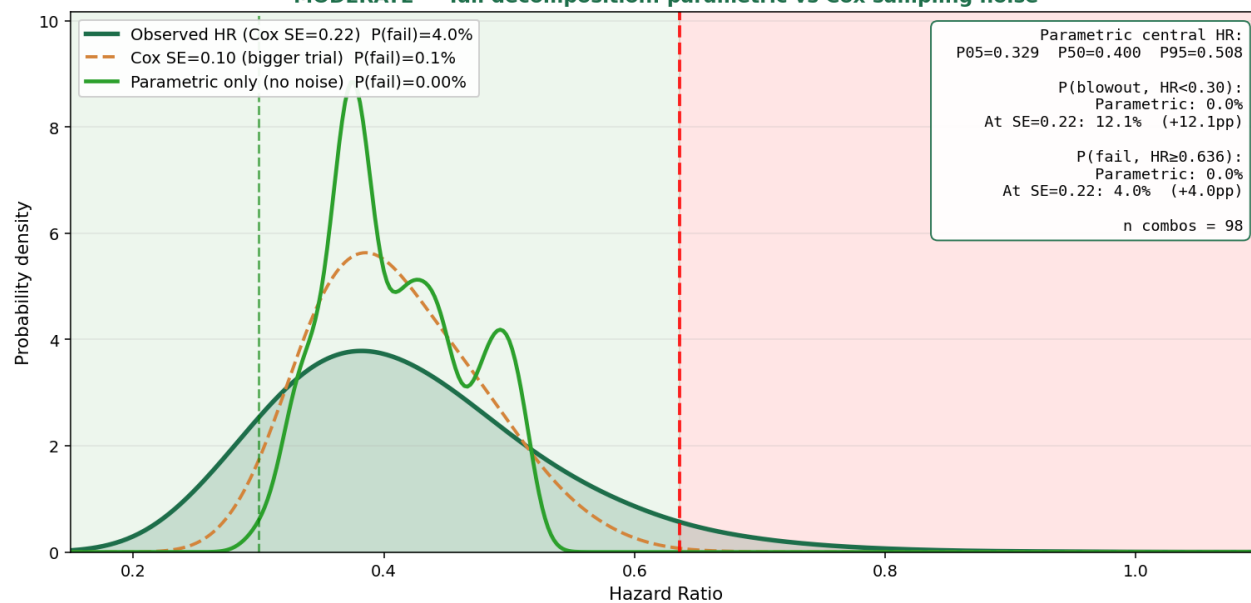
Both tails are sampling-noise artifacts of running a small trial, not genuine biological scenarios. The same noise that creates failure tail risk also creates the chance of a spectacular readout that's better than the true biology.

Tier	Noise level	P(HR < 0.30) blowout	P(HR ≥ 0.636) FAIL
BEDROCK	Parametric (SE=0)	0.0%	5.4%
	Half noise (SE=0.10)	0.6%	8.4%
	Actual (SE=0.22, n=80)	4.6%	15.7%
STRONG	Parametric (SE=0)	0.1%	0.0%
	Half noise (SE=0.10)	2.0%	0.5%
	Actual (SE=0.22, n=80)	9.9%	5.4%
MODERATE	Parametric (SE=0)	0.0%	0.0%
	Half noise (SE=0.10)	2.9%	0.1%
	Actual (SE=0.22, n=80)	12.1%	4.0%

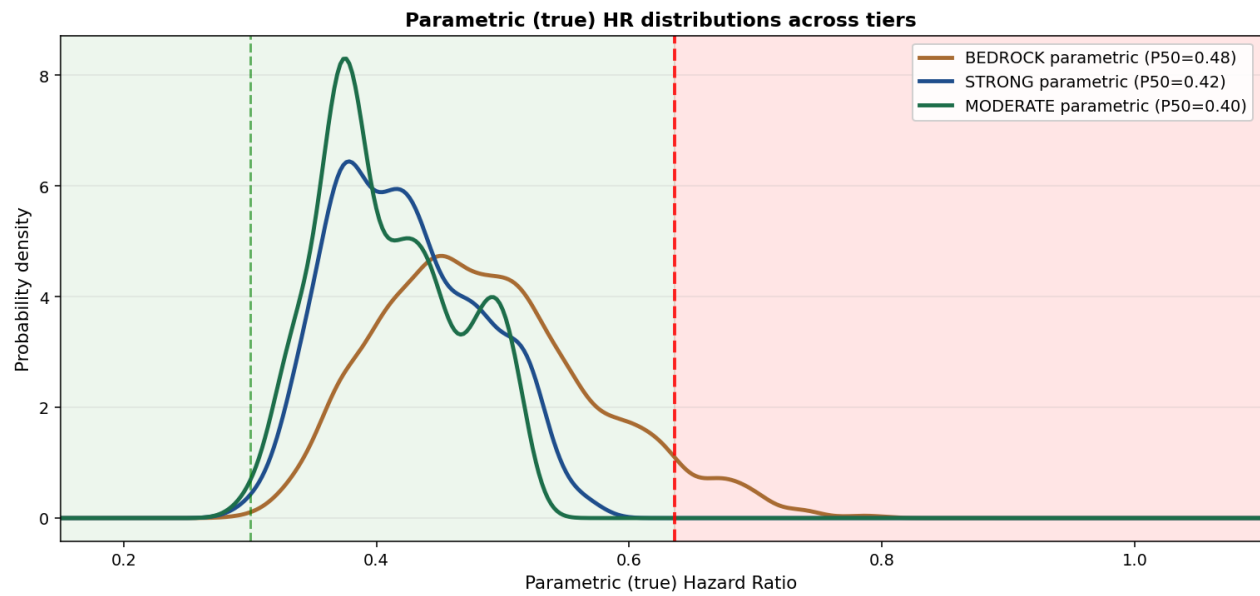
Per-tier tail decomposition charts



MODERATE — Tail decomposition: parametric vs Cox sampling noise



Parametric HR distributions across tiers



Look at how far the parametric distributions sit from $HR = 0.636$. Even BEDROCK's parametric P95 sits at 0.64 — only modestly above threshold. STRONG and MODERATE parametric P95s are well below 0.636. The 4–16 pp observed-HR failure mass across tiers is mostly Cox sampling noise on working biology, not 'the drug doesn't work.'

Important caveats — things NOT in the model

1. **Per-arm event split:** Not disclosed publicly. Could push either direction at readout.
2. **Last 2 events arm-split variance:** Small but mechanically more material now that only 2 events remain.
3. **Open-label bias on OS:** Ambiguous direction. Lead-time / post-progression treatment asymmetry could inflate HR modestly; Hawthorne effect goes BOTH WAYS — investigators may work harder for BAT patients precisely BECAUSE they're not on the vaccine. Net effect approximately neutral. Partly captured by ratio ≥ 0.85 floor.
4. **Sponsor disclosure asymmetry:** Any specific surprises at readout (subgroup analyses, KM curve shape).
5. **WT1 immune escape:** Theoretically possible if cured AML clones lose WT1 expression. No published evidence in CR2 timeframe; mechanism exists.
6. **Selection effect for fitter patients:** REGAL required performance-status criteria; enriched for fitter biology in BOTH arms. Explains BAT ~15m vs historical 6–8m. NOT a confounder for HR (lifts both arms equally).
7. **Allocation asymmetry magnitude:** ratio ≥ 0.85 BEDROCK floor assumes ~1–3 month salvage delay max. Larger asymmetry would lower P(success) further.
8. **Cross-family model misspecification:** Leaky-cure family assumed throughout. Earlier v4 analysis showed all three families converge near MODERATE-equivalent priors.

What would change the headline

Disclosure	Effect on headline
GPS Phase 2 long-term FU showing plateau past 5 years	Strengthens leak prior; STRONG $\rightarrow$ ~92%
GPS Phase 2 late relapses appearing	Weakens leak prior; STRONG drops to ~88%
Per-arm event split disclosed	Resolves arm-split tail (~1 pp either way)
Pool mOS guidance change	Updates BAT posterior; depends on direction
80 events before late June	Modestly bearish (faster events = less plateau signal)
80 events after mid-August	Modestly bullish (slower events = more plateau effect)
Trial conduct disclosure on salvage allocation	Resolves the ratio prior uncertainty

The genuinely actionable empirical question remains GPS Phase 2 long-term follow-up — the one piece of data that would close the 84%–96% range.

Cross-analysis convergence

Multiple independent analyses on the same public data converge in similar windows:

Source	Methodology	Headline P(success)
This framework v6 (leaky-cure, 3-tier)	Bayesian ABC + Cox noise	84% (BEDROCK) – 96% (MODERATE)
Sister Claude #1 (literature-weighted)	Same framework, different calibration	85–88%
Sister Claude #2 (continuous treatment)	Same, continuous leak prior	89–92%
Sister Claude #3 (5-tier with no-harm)	Five-tier comparison	84–96% center ~89%
CW's framework (Cox-corrected)	Point estimates with sampling noise	80–90%
CW's quoted headline	Point estimates without Cox noise	99.99% (overconfident)

Convergence at ~85–96% across multiple independent approaches when methodology is properly applied. The v6 STRONG anchor (94.6%) sits in the upper part of the convergence window; the STRONG–MODERATE midpoint (95%) is the cleanest literature-anchored single number.

Single-line summary

Modal HR **0.37 – 0.44** across the 3-tier range. P(success) **84%** (BEDROCK floor) – **95%** (STRONG center) – **96%** (MODERATE ceiling). Modal readout mid-July 2026, 2 events remaining, ~15–21 months implied survival benefit. The failure tail at the STRONG anchor is ~5% observed — but parametric-only is only 0.0%, meaning ~75% of failure mass is Cox sampling noise on working biology.

Version history

v1 (91%, redundant ratio filter). v2 (94% headline, internally inconsistent). v3 (95% headline, properly Bayesian-weighted, CSV approximation). v4 (96% headline, simulator rerun with 78@m63 + increment constraints). v4.1 (~92% central, fail composition decomposed, EV synthesized). v5 (ratio ≥ 0.85 across all tiers, WEAK dropped). **v6 (current)**: 3-tier restructure (BEDROCK and BEDROCK+ collapsed). Shape ≤ 1.0 throughout (honest given the coarse grid). Ratio progresses $0.85 \rightarrow 0.90 \rightarrow 0.95$. Cure progresses $0.40 \rightarrow 0.45 \rightarrow 0.50$. Leak progresses $0.10 \rightarrow 0.07 \rightarrow 0.05$. Headline range: **84% (BEDROCK) – 96% (MODERATE)**, single-number anchor at STRONG–MODERATE midpoint **~95%**.